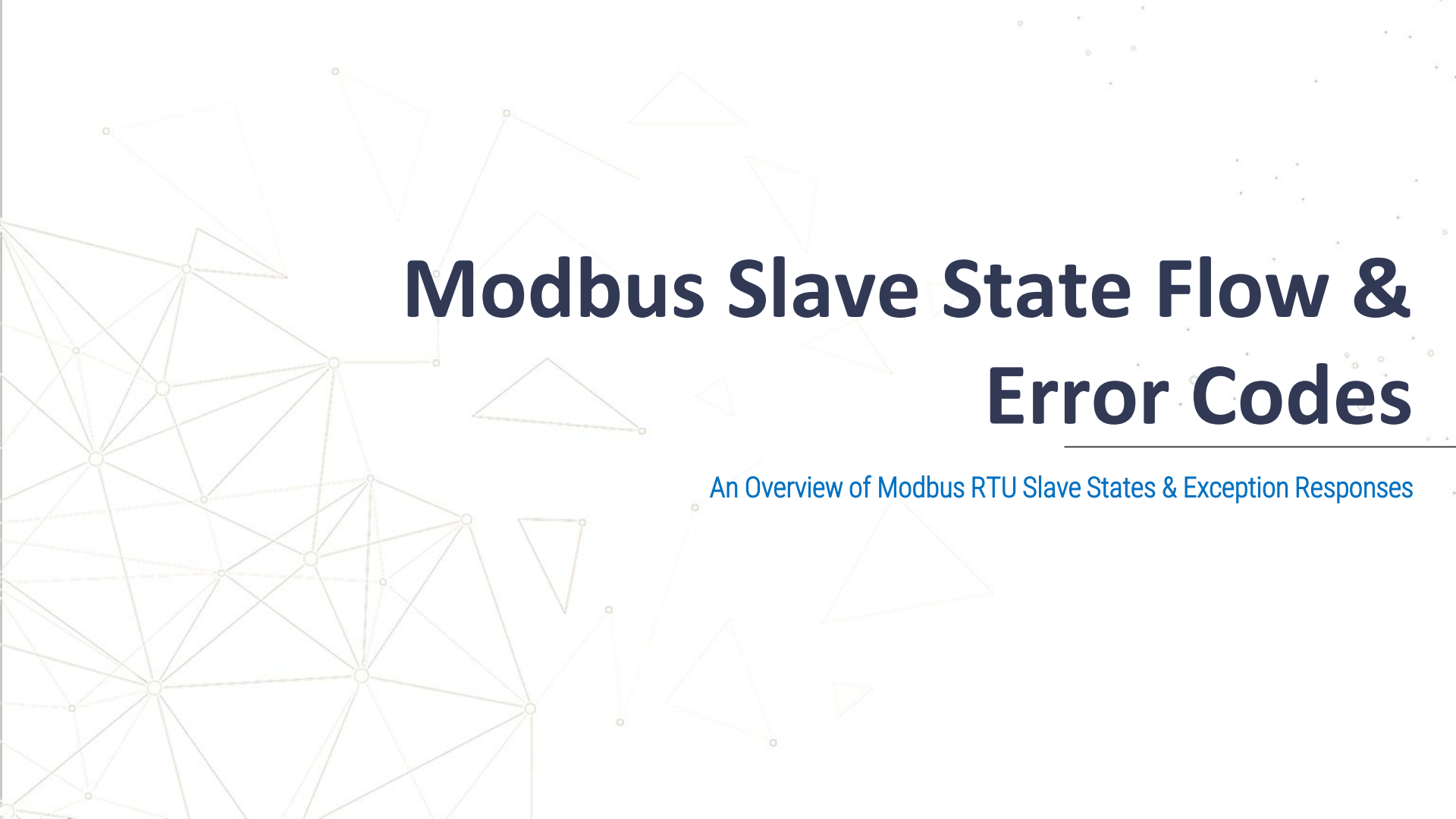


The background features a complex network of thin, light-colored lines and dots, creating a web-like or molecular structure. Scattered throughout are various geometric shapes, including triangles and polygons, some of which are filled with a light yellow or green color. The overall aesthetic is technical and modern.

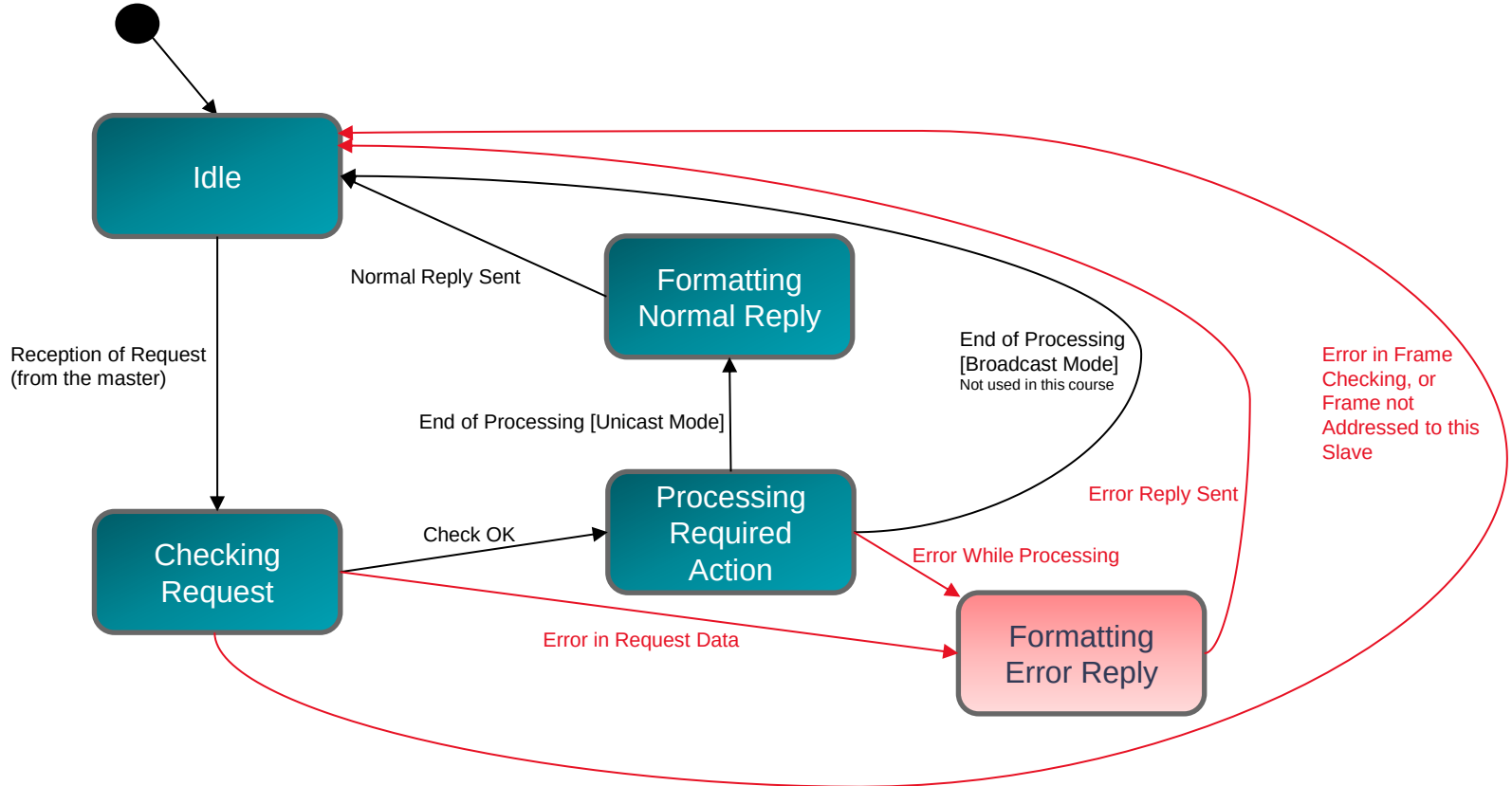
Modbus Slave State Flow & Error Codes

An Overview of Modbus RTU Slave States & Exception Responses

Modbus Slave State Flow & Error Codes

- Modbus Slave State Flow
- Modbus Exception Codes
- Modbus Exception Response Examples
- Modbus Exception Response Examples Continued...

Modbus Slave State Flow



Modbus Exception Codes

Common Modbus Exception Codes

Code	Description	Note
0x01	ILLEGAL FUNCTION	Unsupported function code
0x02	ILLEGAL DATA ADDRESS	Address is out of range.
0x03	ILLEGAL DATA VALUE	Invalid data in the request.
0x04	SLAVE DEVICE FAILURE	Unrecoverable hardware failure.

Note: Additional exception codes which can be implemented, include, 0x05 ACKNOWLEDGE, 0x06 SERVER BUSY, 0x08 MEMORY PARITY ERROR, etc..

Modbus Exception Response Examples

Exception Response Format

Slave Address	Function Code + 0x80	Exception Code	CRC
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- **Illegal Function (0x01):** Master sends Function Code 0x10 to a slave that doesn't support it.

Slave Address	Function Code + 0x80	Exception Code	CRC
07	0x90	0x01	----

- **Illegal Data Address (0x02):** Master requests to read a register outside the valid range.

Slave Address	Function Code + 0x80	Exception Code	CRC
07	0x83	0x02	----

Modbus Exception Response Examples Continued...

Exception Response Format

Slave Address	Function Code + 0x80	Exception Code	CRC
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- **Illegal Data Value (0x03):** Master sends an invalid value for a register.

Slave Address	Function Code + 0x80	Exception Code	CRC
07	0x83	0x03	----

- **Slave Device Failure (0x04):** Master sends a valid request, but the slave encounters an internal error while processing it.

Slave Address	Function Code + 0x80	Exception Code	CRC
07	0x83	0x04	----

The background features a complex network of thin, light-colored lines and small circles, forming a web-like structure. Overlaid on this are several triangles of various sizes and orientations, some of which are filled with a light blue or yellow color. The overall aesthetic is technical and geometric.

Next: Practical Demonstration
